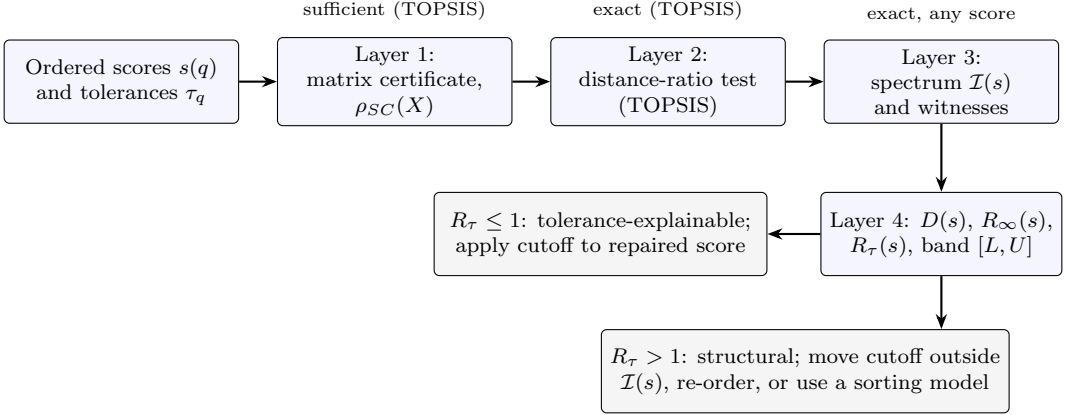

sufficient (TOPSIS)
exact (TOPSIS)
exact, any score
Ordered scores $s(q)$ and tolerances τ_q
Layer 1: matrix certificate, $\rho_{SC}(X)$
Layer 2: distance-ratio test (TOPSIS)
Layer 3: spectrum $\mathcal{I}(s)$ and witnesses
Layer 4: $D(s)$, $R_\infty(s)$, $R_\tau(s)$, band $[L, U]$
$R_\tau \leq 1$: tolerance-explainable; apply cutoff to repaired score
$R_\tau > 1$: structural; move cutoff outside $\mathcal{I}(s)$, re-order, or use a sorting model